S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Diphenhydramine HCl** **Label Claim: 14mg/5ml**	**Method 1:** Reagents: 1. Buffer solution, pH 3.0: 100mL of 0.1N potassium hydrogen phthalate and 40.6mL of 0.1N hydrochloric acid are mixed and diluted to 200mL. OR Dissolve 2.042g of potassium hydrogen phthalate in water and made up to 100mL. 2. Bromocresol green (BCG) solution: dissolve 69.8mg of the dye in 2mL of 0.1N sodium hydroxide and dilute to 1000mL with water. OR 34.9mg of the dye is dissolved in 100mL of 0.1M NaOH and diluted to 500mL with water.

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***Standard solution**:*

20mcg/mL of the substance in water.

Sample solution:

Accurately weigh a quantity of the sample equivalent to 20mg of the compound and dissolve in water (20mcg/mL).

5mL of syrup made to 100mL, 14.3mL of the resulting solution is made up to 100mL.

Procedure**:**

- To 15ml each of the sample and standard solutions, add 60ml of dye solution.
- Add 1 to 2 drops of 1N hydrochloric acid till yellow colour is obtained.
- Add 10ml of buffer solution.
- Extract the complex with three 15ml portion of chloroform.
- Combine the chloroform layer dilute to 50ml.
- Add few crystals of anhydrous Sodium Sulphate and shake.
- Measure the absorbance of each sample and standard solutions at 415nm

		using chloroform as a blank. • Calculate the results by comparison **% Assay of Diphenhydramine HCl** = $\frac{absorbance\ of\ sample}{absorbance\ of\ standard}$ x 100 **Note:** 1. Under acidic conditions, most substances with ternary amino group or quaternary ammonium salts make yellow complexes with the dye, extractable with chloroform.Ammonium chloride, menthol, sodium citrate, paracetamol, saccharine, vitamin C, phenylephrine do not interfere.
3.	**Ammonium Chloride** **Label Claim: 135mg/5ml**	**METHOD 1:** **Procedure:** Dissolve 10mL of the syrup in 20 mL of *water R* and add a mixture of 5 mL of *formaldehyde solution*, previously neutralized to *phenolphthalein solution*, and 20 mL of *water R*. After 1-2 min, titrate slowly with *1 M sodium hydroxide*, using a further 0.2 mL of the same indicator. 1 mL of *1 M sodium hydroxide* is equivalent to 53.49 mg of NH_4Cl. $\% \boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ 53.49}{135}\ x\ 100$

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		REFERENCE: *British Pharmacopoeia, 2024, Ammonium Chloride: Ph. Eur. Monograph 0007.*
4.	**Sodium Citrate** **Label** **Claim: 57mg/5ml**	***Standard Solution:*** Prepare an equivalent of 20mcg/1mL of Sodim Citrate in DM ***Sample Solution:*** Dissolve 5mL of the syrup in water and make it up to 100mL. From the resulting, dilute 3.5mL to 100mL. **Procedure:** To 5mL of each of each solution, add 5mL of 0.1M HCl solution, then add 5ml of 0.1M Ferric Chloride. Make up to 25ml with 0.1M HCl. Measure the absorbance of the solutions at 380nm. **% Assay of Sodium Citrate** = $\frac{absorbance\ of\ sample}{absorbance\ of\ standard}$x 100
5.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette.

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- Open the lid of the prepared media, flame its mouth.
- Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample.
- Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an “S” or “8” shape.
- Allow the media to completely solidify.
- Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a

portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.

- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\text{CFU/mL} = \frac{\text{Total number of colonies obtained x dilution factor}}{\text{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of

		lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats. - The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105** - If the colonies are fused or the whole plate is covered with a single colony, then report as "too numerous to count" (TNTC) and repeat the process of taking the sample at a higher dilution.

*LAF--- Laminar Air Flow

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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